LLM-as-a-Judge for Reliable and Explainable Offline Evaluation in Top-K Recommendation

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Abstract

Recommendation evaluation plays a crucial role in guiding the refinement and deployment of recommender systems. Most existing trials rely on offline evaluation using Top-K metrics computed over holdout user behaviors. However, we identify two fundamental limitations that undermine their ability to deliver reliable and explainable evaluations. Regarding reliability, offline evaluation treat observed user feedback as a proxy of true preferences and enforce rigid ID matching between the proxy and recommendation. In practice, feedback collections are inherently shaped by incomplete and biased item exposure, leading to distorted and unreliable assessments. Regarding explainability, Top-K metrics only establish numerical scores without offering meaningful insights to support them, thereby reinforcing the black-box nature of offline evaluation.

In this paper, we propose a reliable and explainable LLM-as-a-Judge framework for offline recommendation evaluation. To enhance reliability, we introduce a semantic proxy from user textual behaviors to represent their true preferences. This proxy allows for more flexible matching between preferences and recommendations in the semantic space, rather than depending on the holdout feedback. To ensure explainability, the LLM Judge adopts a *reasoning-then-scoring* process to generate relevance judgments along with explicit rationale. Finally, we aggregate the individual scores into global Top-K metrics to quantify overall recommendation quality, and provide justification for each preference hit or miss. Extensive experiments demonstrate that our method achieves significant consistency with unbiased evaluations while exhibiting promising explainability and robustness across LLMs of different scales.

CCS Concepts

• **Information systems** → **Recommender systems**; **Relevance assessment**.

Keywords

Recommender Systems; Large Language Models; Offline Evaluation

ACM Reference Format:

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1 Introduction

Recommender systems enhance user experience by providing personalized recommendation to alleviate information overload. Evaluating recommendation performances is essential for the system design, refinement and deployment. Serving as the widely adopted paradigm in research, offline evaluation assesses recommendation quality by computing common Top-K metrics such as hitting rate, Recall@K, or NDCG@K on holdout feedback collections [37, 45].

Unfortunately, under the widespread presence of missing-not-at-random (MNAR) feedback, offline metrics risk being distorted and thus provide unreliable evaluation results. Ideally, the gold standard for recommendation assessment can be defined as: *to match true preference space of the user.* Offline evaluation simply assumes that collected user feedback is the proxy for the entire preference space. Based on this assumption, offline metrics reflect recommendation quality by performing strict ID matching between the recommended items and the test set. While in practice, user history collections are largely MNAR, which means they only represent a subset of the overall interests due to multiple sources of inherent bias, such as exposure control imposed by the deployed system or user item-viewing behaviors [21, 33]. Consequently, enforcing rigid ID matching on MNAR data fails to account for unobserved interests, resulting in an inevitable deviation from the gold standard.

On the other hand, offline evaluation remains limited in delivering explainable assessments [9]. With growing attention on explainable recommendation, explainability in evaluation is also emphasized to offer meaningful insights into assessment rationality [7, 12]. However, traditional offline metrics are typically confined to establishing numerical performance benchmarks and lack the ability to provide sufficient explanations. Based on these observations, we highlight the fundamental limitation in current offline evaluation: *the absence of reliable and explainable proxy that consistently aligns with the gold standard.*

To move beyond the feedback collection proxy employed in traditional offline evaluation, Large Language Models (LLMs) demonstrate the capability to infer a semantic proxy of the gold standard. With extensive world knowledge and excellent reasoning ability, LLMs can interpret user behavioral histories to capture latent intentions and unobserved interests beyond MNAR feedback. Therefore, the semantic representation offers a more comprehensive summary of user true preferences, enhancing the reliability of the preference proxy. Building on this, the emergence of LLM-as-a-Judge paradigm further promotes explainable evaluation by engaging agents to reason over candidates and align judgments with human-like criteria [10, 14]. Taking together, our approach leverages the semantic understanding and explainability of LLMs to effectively bridge the gap toward reliable and explainable evaluation.

In this paper, we propose a novel LLM-as-a-Judge framework to address the fundamental limitations in offline recommendation evaluation. First, to enhance the reliability in offline evaluation, we

introduce a semantic proxy to represent user preference space, thus moving beyond the reliance on supplementary test sets. Based on this proxy, we establish a semantic matching principle between the proxy and the recommended content, enabling more flexible and logical alignment with user interests. Second, to improve the explainability in offline evaluation, we devise a two-step judging process within the LLM Judge. In the reasoning step, the Judge summaries user intentions from the history to build the semantic proxy, and assesses the relevance between the proxy and recommendation. In the scoring step, the Judge assigns pointwise scores to candidates based on the prior reasoning and outputs explicit rationales. Finally, the judgments are aggregated into global metrics that reflect overall recommendation performance, accompanied by individual reasons explaining each decision. We release our implementation at https://anonymous.4open.science/r/llm4recJudge.

In summary, our contributions are threefold.

- We propose an LLM-as-a-Judge framework that introduces a semantic proxy to represent user true preferences. This framework enables flexible semantic matching between the proxy and recommended content, providing reliable evaluation.
- We devise a reasoning-then-scoring judging process in the LLM Judge to produce assessment scores along with explicit rationales, achieving explainable evaluation.
- We conduct extensive experiments on two real-world datasets, demonstrating the reliability and explainability of the LLM Judge, as well as robustness across LLM backbones of different scales.

2 Methodology

In this section, we first outline the preliminaries of Top-K recommendation and common offline metrics to formulate the offline evaluation problem. Next, we introduce the proposed LLM-as-a-Judge framework for recommendation evaluation.

2.1 Problem Formulation

We study the standard Top-K recommendation task in collaborative filtering, which serves as a fundamental pipeline for building recommender systems [18]. Let $\mathcal{U}$ and $\mathcal{I}$ denote the user and item sets. The collection of user-item interactions is represented as $\mathcal{D} = \{(u, i) \mid u \in \mathcal{U}, i \in \mathcal{I}\}$, where each pair $(u, i) \in \mathcal{D}$ indicates that user u has interacted with item i. For each user u, we denote his historical feedback set as $\mathcal{I}_u^+ = \{i \in \mathcal{I} \mid (u, i) \in \mathcal{D}\}$.

Ideally, a collaborative filtering recommender is trained on the observed interactions $\mathcal{I}_u^+$ to learn a scoring function $s(u, i)$ that estimates estimates a user's preference for each item. During inference, the model computes scores for user u over the set of unseen items $\mathcal{I} \setminus \mathcal{I}_u^+$, and ranks them in descending order of $s(u, i)$ to produce the Top-K recommendation list:

$$\pi_K(u) = \text{Top-K}_{i \in \mathcal{I} \setminus \mathcal{I}_u^+} \; s(u, i), \tag{1}$$

where $\pi_K(u) = \{i_1, \ldots, i_K\}$ denotes the ordered list of the K highest-scoring items. The recommendation model is evaluated by comparing $\pi_K(u)$ with the ground-truth relevant items from the unseen set. We define the set of items that truly match user u's preference as $\mathcal{T}_u \subseteq \mathcal{I} \setminus \mathcal{I}_u^+$.

To evaluate how effectively π_K captures ground-truth preference in $\mathcal{T}$, two widely used offline metrics are employed, which are the recall rate and the normalized discounted cumulative gain (NDCG). Recall@K measures the fraction of relevant items that appear in the Top-K recommendation list:

$$\text{Recall@}K = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \frac{|\pi_K(u) \cap \mathcal{T}_u|}{|\mathcal{T}_u|}. \tag{2}$$

NDCG@K evaluates both the relevance and ranking quality of recommendations by assigning more scores to correct recommendations with higher rankings. The discounted cumulative gain (DCG) is first calculated as:

$$\text{DCG@}K(u) = \sum_{r=1}^{K} \frac{2^{\mathbb{1}\{i_r \in \mathcal{T}_u | i_r \in \pi_K(u)\}} - 1}{\log_2(r+1)}. \tag{3}$$

Then ideal DCG is computed by re-ranking hitted items to the top:

$$\text{iDCG@}K(u) = \sum_{r=1}^{\min(K, |\mathcal{T}_u|)} \frac{1}{\log_2(r+1)}. \tag{4}$$

Finally, NDCG is obtained based on the ratio of DCG to iDCG:

$$\text{NDCG@}K = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \frac{\text{DCG@}K(u)}{\text{iDCG@}K(u)}. \tag{5}$$

In practice, as illustrated in Figure 1 (a), offline evaluation typically uses a holdout subset $\hat{\mathcal{T}}$ from the observed $\mathcal{I}^+$ as a proxy for the unobserved true preference $\mathcal{T}$. Traditional Top-K metrics measure performance through strict ID matching between π_K and $\hat{\mathcal{T}}$, *i.e.,* Metric$(\pi_K, \hat{\mathcal{T}})$, thereby resulting in deviation from the ideal gold standard. To address this issue, we formulate the offline evaluation problem as identifying a proxy $\mathcal{P}$ that yields evaluation results (*e.g.,* Recall@K and NDCG@K) consistent with those computed on $\mathcal{T}$, that is, Metric$(\pi_K, \mathcal{P}) \rightarrow$ Metric$(\pi_K, \mathcal{T})$.

2.2 LLM-as-a-Judge

We introduce the LLM Judge framework in three parts: (1) *Judging Process*, which outlines the procedures for performing pointwise semantic matching and reasoning; (2) *Prompt Construction*, which defines concrete principles to effectively guide the agent to judge; and (3) *Outcome Evaluation*, which aggregates the generated judgments into standard Top-K performance metrics.

2.2.1 Judging Process. We assume that each item is associated with natural language information (*e.g.,* title, attributes, or short description). As shown in Figure 1 (b), given the user historical interactions and a Top-K recommendation list produced by the recommender, the Judge is prompted to the assess the pointwise relevance between the user and each candidate item based on their linguistic features. Central to our LLM Judge is the semantic matching mechanism, which replaces the traditional ID-based matching employed in holdout test evaluations. For clarity of demonstration, we conceptually decompose the judgment into *reasoning* and *scoring* steps, while practically encapsulating both within single LLM query.

In the *reasoning step*, the Judge first summarizes user preferences from the observed textual feedback, forming the semantic proxy that captures their high-level interests. Based on this proxy, the Judge performs semantic matching to identify the relevance between recommendation candidates and the inferred user profile. This semantic-level inference allows the model to rely on linguistic information rather than meaningless item IDs, thereby effectively

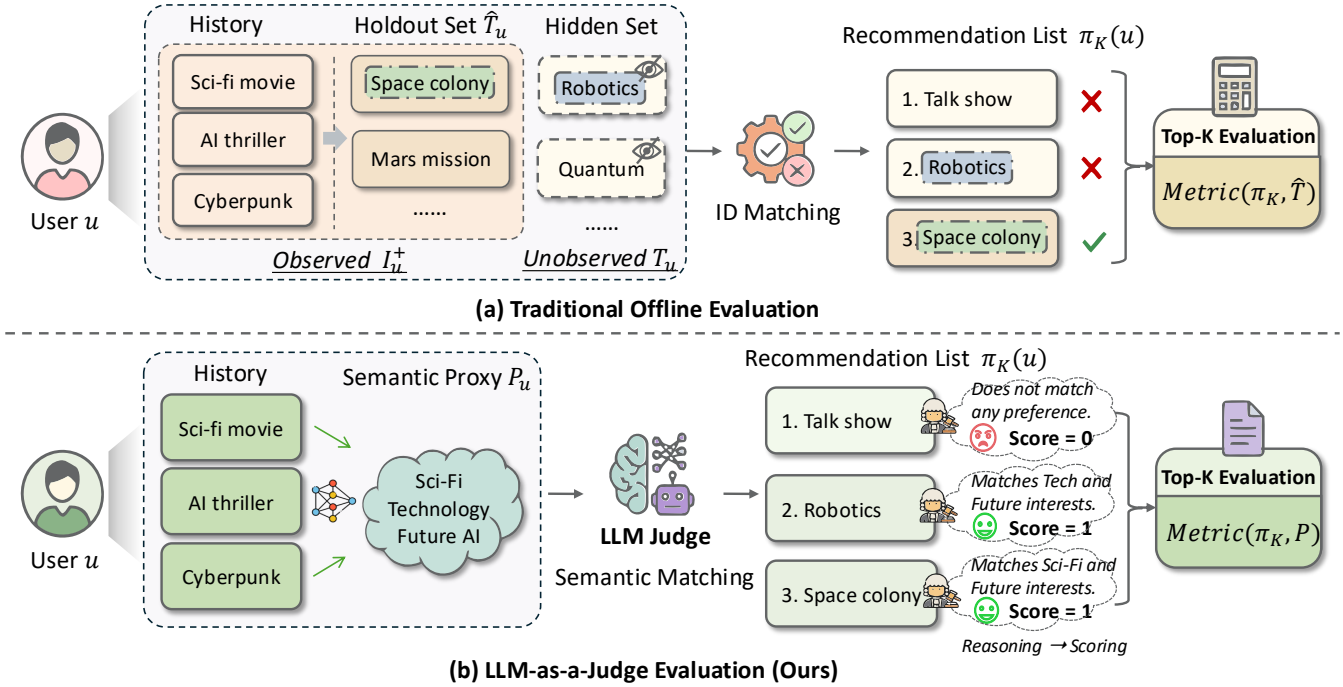


Figure 1: Overview of traditional offline evaluation vs. LLM-as-a-Judge evaluation in Top-K recommendation.

alleviating the MNAR impact inherent in traditional holdout proxies. For each candidate item, the matching judgment involves a chain-of-thought reasoning to cite key evidence that leads to accept or reject decision. The reasoning outcome is a single-sentence rationale explaining why the item aligns (or mismatches) the underlying preferences. This stage guides the Judge to ground its assessment on observable textual cues, instead of directly guessing the score.

In the *scoring step*, the Judge assigns a pointwise relevance score to each candidate item conditioned on the preceding reasoning rationale. The score indicates whether the candidate is relevant to the inferred user preferences (1 for relevant, 0 for irrelevant). Along with the score, rationale of the judgment is also reported to ensure transparent and interpretable evaluation outcome.

2.2.2 Prompt Construction. Our prompt is designed to implement a strict pointwise relevance evaluator. To ensure generalizability, the prompt template standardizes only the general evaluation procedure, while task-specific details are provided through customizable input fields to maintain scalability across various recommendation tasks. As shown in Figure 2, the prompt template explicitly specifies three key aspects in the evaluation process: *basic principles*, *input fields*, and *output format*.

Basic principles. We define two fundamental rules to guide the entire judgment process, (1) the reasoning and scoring principle, and (2) the reject-by-default policy. The first rule regulates the judge to follow our proposed two-step judgment procedure, while the reject-by-default policy enables a conservative LLM judge to mitigate the over-approve tendency in LLMs. Given that LLMs often pursue divergent associations and may exhibit hallucinations [19], we explicitly constrain the judge to always reject the candidate unless the provided evidence clearly supports a perfect match. This rule is therefore essential for filtering out ambiguous candidates and ensuring an accurate evaluation process.

Input fields. The input fields are serialized into a unified template consisting of (1) an instruction block that defines the task and matching criteria, (2) the textual fields representing the user historical interactions and the recommended candidate information. In the instruction block, the current recommendation task requires customized definition. The matching criteria are originally task-agnostic with a general description of the perfect matching situation, but we also allow flexible customization to better fit the corresponding task. This modular design allows the same judging logic to transfer across different recommendation scenarios by simply adjusting the instruction block while keeping the remaining template and rubric unchanged.

Output format. At last, the judge is required to output a binary relevance score (0 or 1) along with a single-sentence rationale explaining the judgment. The binary scores are then aggregated into global metrics to reflect the overall recommendation quality, while their corresponding explanations are retained for further qualitative analysis or error diagnosis to better understand the Judge behavior and reasoning patterns.

2.2.3 Outcome Evaluation. Since the judgment score takes binary value, we follow typical offline evaluation protocol to aggregate the LLM evaluation outcomes with standard Top-K recommendation metrics. In this study, we select the representative Recall@K and NDCG@K as primary evaluation metrics, while other list-level metrics remain compatible with our framework. We make minor adjustments to adapt the two metrics to our evaluation framework.

Prompt Template

SYSTEM PROMPT

You are a strict and impartial recommendation judge for [INPUT: TASK DEFINITION].
Your goal is to determine if a candidate item is a perfect match for the user based on user history.

BASIC PRINCIPLES

1. Reasoning-then-scoring: First infer user preferences from the history, then think about the reason that leads to matching or mismatching, finally make your judgment based on the reasoning.
2. Reject-by-default: Your starting position is REJECT (0). Remember most recommendations are mediocre, you only approve it if the evidence is undeniable.

DECISION FRAMEWORK

0 : REJECT (Default)
- [INPUT: MATCHING CRITERIA]

1 : HIT (Strict Match)
- [INPUT: MATCHING CRITERIA]

OUTPUT FORMAT

You MUST output a JSON object with 'reason' and 'score' fields. The 'reason' should be a single, concise sentence in English.

USER PROMPT

User history: [INPUT: HISTORY]
Candidate item to evaluation: [INPUT: CANDIDATE]

Figure 2: Evaluation Prompt applied in our LLM Judge. Basic principles and output format highlighted in boldface are fixed regions, while the input fields shaded in gray can be customized for different tasks.

Formally, given the Top-K list $\pi_K(u) = \{i_1, \ldots, i_K\}$, and the LLM judge outcome $y_{u,i_r} \in \{0, 1\}$ for each rank r. $\text{NDCG}_{\text{Judge}}@K$ is computed in the standard way by treating y_{u,i_r} as the relevance signal in DCG. For Recall@K, since our protocol does not rely on the holdout test set to represent $\mathcal{T}_u$, and thus $|\mathcal{T}_u|$ is unavailable. Accordingly, we normalize the number of judged hits by K:

$$\text{Recall}_{\text{Judge}}@K = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \frac{1}{K} \sum_{r=1}^{K} y_{u,i_r}. \tag{6}$$

3 Experiment

In this section, we conduct extensive experiments to validate the effectiveness of our LLM Judge in achieving reliable and explainable evaluation for Top-K recommendation. We aim to answer the following research questions:

- **RQ1:** Can the LLM Judge be a reliable proxy for the unbiased offline evaluation?
- **RQ2:** Can the LLM Judge generate logically sound and persuasive explanations for its judgment?
- **RQ3:** How robust are the evaluation results using LLM Judges with different model scales?

3.1 Evaluation Pipeline

3.1.1 Data Overview. We employ two real-world datasets, Kuairec[1] [13] and Coat[2] [30], both of which are widely used for unbiased evaluation in recommender systems. Kuairec is a video recommendation dataset containing 12.5 million interactions between 7.1k users and 10.7k items. Coat dataset records 6.5k ratings provided by 290 users on 300 clothes from a shopping platform. For model training, both datasets adopt the same MNAR feedback as most recommendation datasets. While for testing, they support unbiased evaluation with dedicated test sets. Specifically, Kuairec provides a fully observed test subset containing 1.4k users and 3.3k items, where each user has left feedback on every item in the subset. Coat offers an unbiased test set in which all users have rated 16 randomly exposed items. Among available real-word datasets, only Kuairec, Coat, and Yahoo!R3 [27] provide unbiased offline test data. We exclude Yahoo!R3 as it contains only item IDs without textual information, remaining inapplicable to our setting. We provide the textual information details of Kuairec and Coat in Appendix A.1.

3.1.2 Baseline Construction. We design two pipelines to establish the lower and upper bounds of evaluation quality, namely normal testing and unbiased testing. We split the MNAR feedback of each dataset into training and validation sets with a 7:3 ratio, and train recommendation models on the training set. In normal testing, model performances are evaluated by the validation set, which represents a lower bound of evaluation quality due to incomplete item exposure. In contrast, unbiased testing assesses the same trained models on the unbiased test sets, which are collected under full item exposure, thereby serving as the evaluation upper bound. To enable unbiased testing, we binarize the fully exposed test data into positive and negative samples, defining positives as videos watched at least twice in Kuairec and ratings above 3 in Coat dataset. We then treat the unbiased testing as the ground truth, and consider both normal testing and LLM Judge as proxy evaluation schemes that approximate this ground truth. Among them, normal testing serves as the baseline for comparison with the LLM Judge. Finally, to characterize the evaluation landscapes of each proxy and the ground truth, we instantiate them as performance ranking lists of recommendation models derived from their respective evaluations.

3.2 Experimental Settings

3.2.1 Recommendation Models. To build the model ranking list, we implement a diverse set of representative collaborative filtering models. They include matrix factorization methods: MF [26] and NeuMF [16]; graph neural network models: NGCF [38], LR-GCCF [6], and LightGCN [15]; constrastive learning model: SimGCL [44], and graph-based variants that improve high-order connectivity modeling: MGCCF [32], GTN [11], and CAGED [29]. Rather than identifying an optimal recommender in the list, we focus on the relative model rankings under the two proxy schemes, and assess their consistencies with the unbiased ordering to reflect evaluation reliability. We attach model implementation details in Appendix A.2.

3.2.2 Evaluation Metrics. We employ a variety of evaluation metrics to examine the reliability and explainability of our LLM Judge

[1] https://kuairec.com/
[2] https://www.cs.cornell.edu/~schnabts/mnar/

Table 1: Consistency performance of normal testing and LLM Judge evaluation. The best consistency results are highlighted in boldface. Gain denotes the performance improvement over normal testing. Superscript * indicates the statistically significant consistency with the ground truth (*i.e.*, two-sided t-test with $p < 0.05$).

Dataset	Metric	Normal Testing	LLM Judge	Gain	Normal Testing	LLM Judge	Gain	Normal Testing	LLM Judge	Gain
Kuairec	Top-K	Recall@20			Recall@50			Recall@100		
	Pearson Correlation	35.56%	**87.62%***	+52.07%	31.93%	**79.79%***	+47.86%	27.88%	**76.71%***	+48.83%
	Spearman's Rank	26.19%	**85.71%***	+59.52%	19.05%	**64.29%**	+45.24%	16.67%	**79.42%***	+62.75%
	Kendall's τ	28.57%	**71.43%***	+42.86%	28.57%	**42.86%**	+14.29%	21.43%	**61.83%***	+40.40%
	Top-K	NDCG@20			NDCG@50			NDCG@100		
	Pearson Correlation	2.65%	**62.61%**	+59.96%	-2.88%	**70.79%***	+73.66%	2.43%	**71.35%***	+68.92%
	Spearman's Rank	11.90%	**71.43%***	+59.52%	16.67%	**78.57%***	+61.90%	16.67%	**80.95%***	+64.29%
	Kendall's τ	14.29%	**50.00%**	+35.71%	21.43%	**64.29%***	+42.86%	21.43%	**64.29%***	+42.86%
Coat	Top-K	Recall@3			Recall@5			Recall@10		
	Pearson Correlation	38.15%	**87.41%***	+49.25%	34.06%	**85.70%***	+51.64%	-8.91%	**76.20%***	+85.11%
	Spearman's Rank	23.81%	**92.86%***	+69.05%	9.52%	**52.41%**	+42.89%	-16.67%	**38.10%**	+54.76%
	Kendall's τ	21.43%	**78.57%***	+57.14%	7.14%	**40.01%**	+32.86%	-7.14%	**21.43%**	+28.57%
	Top-K	NDCG@3			NDCG@5			NDCG@10		
	Pearson Correlation	-77.57%	**73.86%***	+151.43%	-48.31%	**85.61%***	+133.93%	-30.12%	**82.72%***	+112.85%
	Spearman's Rank	-76.19%	**80.95%***	+157.14%	-47.62%	**76.19%***	+123.81%	-45.24%	**83.33%***	+128.57%
	Kendall's τ	-57.14%	**64.29%***	+121.43%	-28.57%	**57.14%***	+85.71%	-35.71%	**71.43%***	+107.14%

separately. To evaluate the reliability of proxy evaluation, we leverage four statistical metrics as follows:

- **Pearson Correlation:** measures the consistency of score gaps between neighboring ranks.
- **Spearman's Rank:** evaluates the consistency of overall ordering positions between two ranked lists, irrespective of the score scale.
- **Kendall's τ:** quantifies the consistency of pairwise order relations between all entry pairs across the two lists.
- **Mean Absolute Error (MAE):** computes the average absolute difference between the proxy and unbiased evaluation scores after normalization.

Among these, the first three are correlation-based metrics taking values within $[-1, 1]$, where 1 indicates perfect consistency, 0 denotes no correlation, and -1 reflects complete inverse correlation. In contrast, MAE serves as an accuracy-based metric that captures the magnitude of deviation between two normalized evaluation scores. Since we summarize recommendation performances in Top-K metrics (*i.e.,* Recall@K and NDCG@K), for comprehensive comparison, evaluations are conducted on recommendation lists with various lengths K. According to the sample size in each dataset, we employ $K \in \{20, 50, 100\}$ in Kuairec, and $K \in \{3, 5, 10\}$ in Coat.

To assess the explainability of the LLM Judge, we design explanatory rubrics from three complementary dimensions:

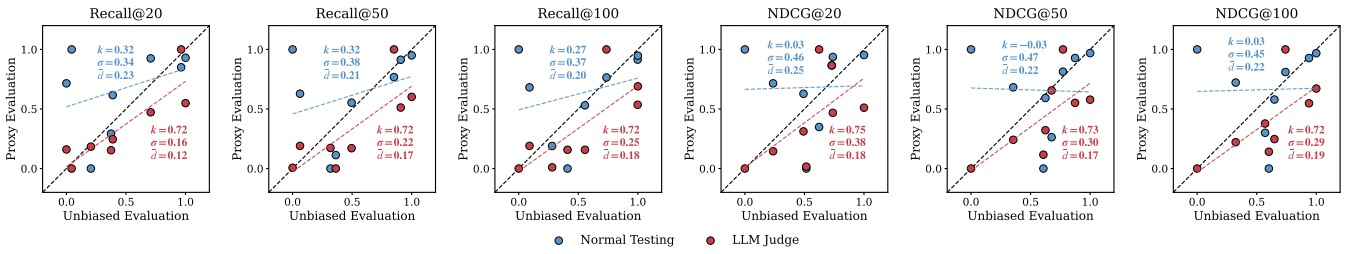
- **Coherence:** whether the reasoning presented by the LLM Judge follows a logically consistent chain of thought that justifies its final decision.
- **Faithfulness:** whether the evidence mentioned in the explanation is genuinely supported by historical data rather than being hallucinated.
- **Persuasiveness:** whether the rationale can convince human evaluators to its judgment.

3.2.3 LLM Judge Backbones. To examine the evaluation robustness of the LLM Judge, we experiment with a series of LLMs with varying scales to build the Judge, including Qwen3-4B-Instruct-2507 [42], Qwen3-8B, Qwen3-14B, Qwen3-30B-A3B, and DeepSeek-V3.2 [8]. As we mainly focus on investigating the effect of LLM capacity on the Judge performance, we do not introduce more models from other LLM families for comparison. In the main experiments, Qwen3-30B-A3B is adopted as the default backbone to balance evaluation performance and computational overhead. For all backbones, the temperature is fixed at 0 to ensure deterministic outputs.

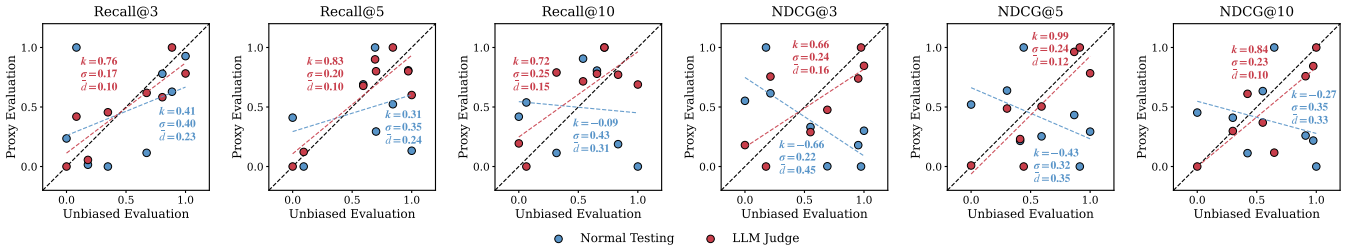
3.3 Study of Evaluation Reliability (RQ1)

To validate the reliability of the LLM Judge as a proxy for unbiased offline evaluation, we compare the Judge performance with normal testing from two perspectives. From a relative ranking perspective, we investigate whether the Judge can accurately recover the relative performance order of recommendation models by measuring the ranking consistency between proxy evaluation and unbiased testing across multiple Top-K metrics. From an absolute score perspective, we examine whether the Judge can produce consistent performance scores by directly comparing the normalized proxy scores with the unbiased scores at the model level and analyzing their overall trends. We present detailed observations for the two aspects below.

Ranking analysis. In Table 1, we report the ranking consistency between each proxy evaluation scheme and the unbiased testing under different Top-K metrics, measured by Pearson correlation, Spearman's rank, and Kendall's τ. On both datasets, normal testing exhibits low consistencies across all Top-K metrics, indicating that MNAR feedback substantially distorts both the relative rankings and the score gaps among recommendation models. As a result, normal testing fails to align with unbiased testing, leading to unreliable performance rankings. In contrast, the LLM Judge consistently achieves strong correlations across all Top-K metrics, showing solid superiority over normal testing in maintaining ranking consistency with the ground truth. Notably, on the NDCG evaluations of Coat,



(a) Kuairec



(b) Coat

Figure 3: Scatter plot between proxy evaluation and unbiased evaluation. Diagonal line $y = x$ represents absolute absence of discrepancy between two evaluation types. Linear regression statistics (k denotes slope, σ represents standard deviation, and $\bar{d}$ indicates MAE) of normal testing and LLM Judge are plotted to illustrate the accuracy trend.

the Judge not only improves the consistency values but also reverses strong negative correlations of normal testing into positive agreement. Furthermore, significance tests confirm that the consistency achieved by the Judge is statistically significant rather than incidental. Therefore, the LLM Judge demonstrates promising capability to recover the relative model rankings much closer to the unbiased testing results.

Score analysis. Figure 3 illustrates the score-level agreement between proxy evaluations and unbiased testing, where each point represents a recommendation model with normalized performance scores in $[0, 1]$. We perform regression analysis for both normal testing and LLM Judge, where the slope k reflects how well each proxy preserves relative score gaps (ideal value $k = 1$) and the standard deviation σ indicates evaluation stability. $\bar{d}$, representing the MAE value, is computed as the average distance of points from the diagonal $y = x$ to quantify the absolute discrepancy from unbiased scores. In normal testing, points are widely dispersed with noticeable deviation from the diagonal line, exhibiting substantial bias in score alignment. This pattern is further supported by the regression statistics, where the slope k generally deviates from the ideal value 1, and the standard deviation σ indicates greater instability in evaluation. Additionally, the MAE result $\bar{d}$ reveals larger differences from the ground-truth scores. In contrast, under the LLM Judge evaluation, performance points cluster more closely around the diagonal, demonstrating a clearer linear trend. Correspondingly, the slopes approach 1 with both lower σ and $\bar{d}$, suggesting that the Judge not only better preserves the relative score differences among models but also provides more accurate absolute performance estimates.

These findings confirm that the LLM Judge attains both ranking consistency as well as score-level agreement with unbiased testing. Consequently, the Judge effectively improves the reliability of offline evaluation over normal testing.

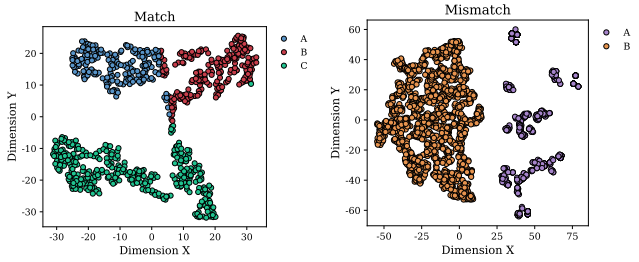
3.4 Study of Evaluation Explainability (RQ2)

To investigate the evaluation explainability of the LLM Judge, we analyze the judgment rationales with both quantitative and qualitative approaches. In the quantitative analysis, we invite human experts to assess the quality of the rationales along the three key dimensions: coherence, faithfulness, and persuasiveness (*cf.* section 3.2.2). In the qualitative analysis, we further conduct the case study on rationale patterns to explore the types of reasoning that the Judge tends to produce.

Rationale quality assessment. We first categorize the Judge outcomes into *Match* and *Mismatch* groups based on the Judge decision, as the two types of reason differ significantly. From each group, we randomly sample 100 judgment rationales, and engage three human evaluators to manually annotate each rationale with binary scores across the three dimensions. The average human evaluation results are summarized in Table 2. For Coat, both *Match* and *Mismatch* groups achieve comparable scores, suggesting that the Judge consistently produces coherent and well-grounded explanations for its persuasive decisions. For Kuairec with larger item space and noisier textual information, *Match* rationales clearly outperform *Mismatch* across all three criteria. This indicates that, under more complex task, the Judge concentrates its reasoning efforts on identifying logical connections that support *Match* judgments. In general, overall performance in both datasets demonstrate solid

Table 2: Explainability analysis of the judgment rationales. Superscript * indicates high inter-rater consistency (*i.e.*, Fleiss' $\kappa \geq 0.75$).

Dataset	Judgment	Coherence	Faithfulness	Persuasiveness
Kuairec	Match	84.67%*	88.33%*	86.67%*
	Mismatch	69.33%*	75.33%*	72.33%*
	Overall	77.00%*	81.83%*	79.50%*
Coat	Match	86.67%*	81.33%*	80.67%*
	Mismatch	84.00%*	85.33%*	83.67%*
	Overall	85.33%*	83.33%*	82.17%*



(a) Match (b) Mismatch

Figure 4: Clustering results of the judgment rationales after Principal Component Analysis (PCA) dimension reduction.

explainability in the judgment rationales, effectively addressing the absence of explainability in traditional offline evaluation.

Case study of rationale pattern. We design the case study on Coat dataset to visualize the rationale landscape generated by the Judge. Specifically, we encode the rationales into sentence embeddings using the BGE-M3 tokenizer [5]. Then we apply K-Means clustering [20] to the *Match* and *Mismatch* groups seperately, and determine the optimal number of clusters using the elbow method. The resulting clusters are visualized in Figure 4, where each cluster represents a distinct type of reasoning to the judgments. We summarize the representative reason example that closest to each cluster center in Table 3. In the *Match* group, there are three major categories of reasons. Cluster A corresponds to logical extension, where the Judge generalizes user preferences to infer broader matches. Cluster B reflects direct matching to support decision through explicit attribute overlap. Cluster C conforms to cross matching, where the Judge links elements in the candidate to multiple related signals in the history. For the *Mismatch* group, Cluster A is primarily associated with gender discrepancies, while Cluster B covers various attribute-level conflicts with user preferences. Overall, these patterns further demonstrate the Judge capability to produce structured and informative rationales, complementing the quantitative findings to establish comprehensive evaluation explainability.

3.5 Study of Evaluation Robustness (RQ3)

To examine whether the evaluation reliability remains robust across different LLM backbone scales, we implement the Judge using a series of LLMs with varying model sizes. For each backbone, we keep the evaluation protocol and prompts unchanged to obtain the corresponding performance evaluation list. We then measure the alignment between each list and unbiased testing results using the same metrics as in the main experiments.

Figure 5 summarizes the evaluation quality of each backbone in terms of Recall@5 and NDCG@5 on the Coat dataset. In each subfigure, the dashed horizontal line represents the normal testing benchmark, serving as a reference for consistency under the conventional offline evaluation setting. Across both Recall and NDCG, the LLM Judge demonstrates strong consistency with the ground truth for all backbones and clearly surpasses the baseline benchmark. In particular, correlation-based measures such as Pearson, Spearman, and Kendall remain significant across model scales, indicating that the Judge reliably preserves the relative ranking of models regardless of underlying LLM scales. Additionally, the MAE remains consistently lower than the baseline, reflecting stable and superior score-level agreement with the unbiased results.

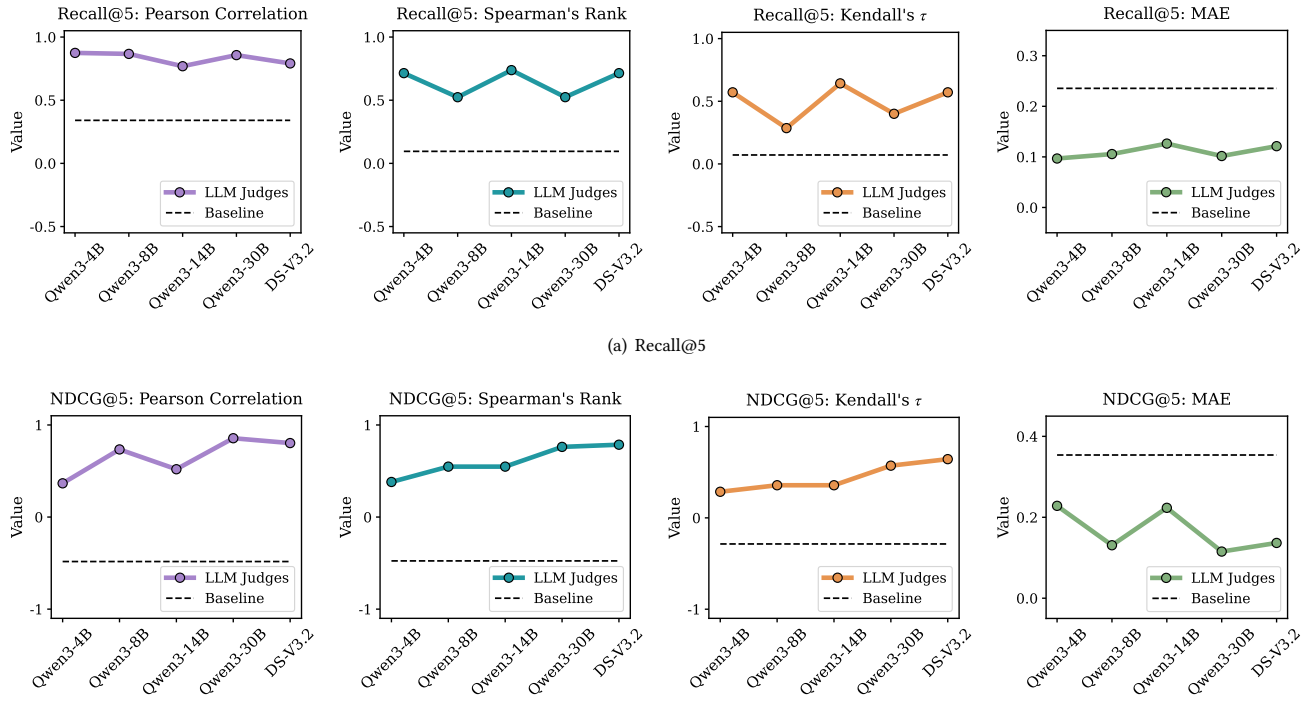
In summary, the LLM Judge consistently maintains high-quality evaluation performance across varying scales, demonstrating strong robustness to backbone selection. This finding suggests that the proposed Judge framework is not exclusively dependent on specific LLM capacity. On the contrary, the LLM Judge framework provides a reliable proxy evaluation that can be flexibly deployed under different computational constraints.

Table 3: Reason examples selected from the cluster centers.

Judgment	Cluster	Reason Example
Match	A	The candidate jacket type is **packable**, which aligns with the user's established preference for **lightweight, functional outerwear**, indicating a strong stylistic and functional fit.
	B	The candidate matches the user's established preference for **packable jackets** and **brown color**, supported by prior purchases of a packable jacket and multiple brown-colored items.
	C	The candidate is a **black parka**, which aligns with the user's established preference for parkas (purchased in brown, gray, and other colors) and black (seen in insulated and motorcycle jackets).
Mismatch	A	The candidate is for **men**, while the user's entire purchase history consists of **women's jackets**, indicating a gender mismatch.
	B	The candidate is a track jacket in **gray**, which does not align with the user's strong preference for **black** and other specific colors like olive, green, and navy in similar jacket types, and gray is not represented in the history.

4 Related Work

Evaluation in recommender systems. Existing approaches generally follow two paradigms: online and offline methods. Online evaluation, such as A/B testing and interleaving experiments, directly measures user responses in real-world deployments, thus providing a faithful reflection of system utility [2, 17]. To mitigate the heavy dependence on live traffic, off-policy evaluation (OPE) is applied to



(a) Recall@5

(b) NDCG@5

Figure 5: Evaluation performance on Recall@5 and NDCG@5 of the Coat dataset. Dashed baseline represents the benchmark of normal testing.

estimate online performance using historical data sampled from the previously deployed logging policy [23, 28]. Nevertheless, in many research settings, real-time interaction environments and logging policies are often unavailable or costly to obtain. Hence, offline evaluation remains the predominant paradigm for assessing recommender systems [1, 3, 4]. In offline setting, substantial efforts have been devoted to addressing the problem of MNAR feedback. For instance, studies in causal inference utilize propensity-weighting techniques to model item exposure, approximating a missing-at-random landscape for unbiased evaluation [22, 25, 31, 43]. However, this approach stays in the rigid ID matching manner to only focus on debiasing exposed items, while neglecting false negative samples. Another direction lies in achieving online-offline alignment, which employs user dynamics from online experiments to guide offline metric design [24, 40]. Yet, as previously discussed, this method is largely constrained in applicability.

LLM-as-a-Judge. The emergence of LLM-as-a-Judge enables scalable, explainable, and human-aligned assessment across a wide range of tasks [34, 39, 47]. This paradigm leverages the reasoning and world knowledge of LLMs to conduct nuanced judgments, often showing strong agreement with human evaluators [14]. In the field of information retrieval, LLM judges are initially investigated as the substitute to human annotators for relevance assessment [35, 41]. Recently, this line of research has expanded into recommender systems. For example, Zhang et al. [46] employ LLM evaluators to assess explanations generated by explainable recommendation models, while Tokutake et al. [36] introduce offline serendipity evaluation using LLMs. These studies focus on specific aspects of recommendation, differing from our broader perspective. Notably, Fabbri et al. [9] propose a profile-based LLM judge for podcast recommendation evaluation. However, their approach primarily relies on well-structured user podcast profiles, which restricts its all-scenario adaptability. In contrast, our judge paradigm transforms user historical traces into semantic preference representations, ensuring strong generalizability across various recommendation tasks.

5 Conclusion and Future Work

In this paper, we present the LLM-as-a-Judge framework to offer reliable and explainable offline evaluation in Top-K recommendation. To achieve reliable evaluation, we introduce a semantic proxy that summarizes user preferences from behavioral contexts, and replace rigid ID matching on holdout feedback with flexible semantic matching between the preference space and recommended items. To realize explainable evaluation, we adopt a reasoning-then-scoring judging process to generate pointwise relevance scores together with informative rationales. Extensive experiments demonstrate that the LLM Judge achieves solid reliability, explainability, and robustness in evaluation. Our future work will investigate the extension of this evaluation paradigm beyond Top-K recommendation to other mainstream recommendation scenarios, such as sequential, cross-domain, and real-time recommendation.

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Table 4: Input statistics of user histories used by the Judge.

Statistic	Kuairec	Coat
# Users	1,411	290
Avg. history length	49.24	15.51
Median history length	50	16
History length range	[26, 50]	[14, 16]
Avg. caption length (tokens)	71.94	62.99
Caption length range (tokens)	[21, 2023]	[58, 69]

Example: LLM Judge Input (Coat)

USER INPUT

User history (all historical interactions):

- gender: women, jackettype: cropped, color: black
- gender: women, jackettype: trench, color: other
- gender: women, jackettype: fleece, color: black
- gender: women, jackettype: fleece, color: gray
- gender: women, jackettype: other, color: blue
- gender: women, jackettype: trench, color: gray
- gender: women, jackettype: cropped, color: green
- gender: women, jackettype: rain, color: purple
- gender: women, jackettype: pea, color: black
- ··· (remaining historical items)

Candidate item to evaluate: gender: women, jackettype: parka, color: black

Figure 6: Example of a complete LLM Judge input on the Coat dataset. Only the textual fields actually used by the Judge are shown. All historical interactions are included without truncation due to the short, structured attribute descriptions.

A Additional Details

This appendix provides supplementary details that further clarify the proposed LLM Judge and support the reproducibility of the experimental results reported in the main paper.

A.1 Input Context Details

To make the evaluation procedure fully transparent, we detail the exact textual inputs provided to the LLM Judge for each dataset, including the input format, field composition, and basic statistics of user histories.

Table 5 summarizes the dataset-specific textual fields consumed by the LLM Judge. Across both Kuairec and Coat, the Judge operates exclusively on human-readable textual information and does not rely on item identifiers, interaction labels, or model-generated features. Importantly, all textual fields are provided in full without truncation or heuristic compression.

For Kuairec, user histories and candidate items are represented using raw video-related textual metadata, including user-written captions, topic tags, and hierarchical category names. These fields are preserved verbatim to reflect the natural variability and richness of real-world user-generated content. For Coat, both user histories and candidate items are expressed using short, structured attribute descriptions, consisting of gender, jacket type, and color. Due to their compact and fixed-format nature, all attributes are included directly without modification.

Figure 6 illustrates a complete example of an LLM Judge input from the Coat dataset. The example explicitly shows how a user's historical interactions are formatted as an unordered list of attribute-level descriptions and how a candidate item is presented using the same attribute schema. Only the textual fields actually consumed by the LLM Judge are shown. Because Coat histories are short and bounded, all historical interactions are included in full.

To further characterize the input context provided to the Judge, Table 4 reports basic statistics of user histories for both datasets. The statistics highlight the contrasting input characteristics: Kuairec features long and diverse histories with highly variable caption lengths, whereas Coat contains short, tightly bounded histories composed of compact attribute descriptions. These differences motivate our unified design choice of supplying the LLM Judge with complete textual histories in both cases, ensuring that preference inference is grounded in the full available user context rather than partial or truncated observations.

A.2 Recommendation Model Details

For reproducibility, we provide the hyper-parameter configurations in the recommendation model training. The trained models are then utilized to generate recommendation lists for evaluation under both traditional offline metrics and the LLM Judge. The general hyper-parameter settings are summarized in Table 6, while for other model-specific hyper-parameters, we follow their optimal settings suggested in each paper.

Table 5: Field-level textual inputs used by the LLM Judge for each dataset. Only human-readable fields are used, and all fields are provided in full without truncation.

Dataset	Field / Attribute	Type	Description / Example
Kuairec	manual_cover_text	Caption	Cover text added by the video author (e.g., "Discovered by the little cutie")
	caption	Caption	Video title or description written by the author
	topic_tag	Caption	Topic tags associated with the video (e.g., [Pomeranian Shunsuke, Live Streaming, Cute Pets Arrival])
	first_level_category_name	Category	First-level category name (e.g., *Pet*)
	second_level_category_name	Category	Second-level category name (e.g., *Pet Daily*)
	third_level_category_name	Category	Third-level category name (e.g., *Pet Dog*)
Coat	gender	Attribute	Target gender of the jacket (e.g., women)
	jackettype	Attribute	Jacket type such as cropped, trench, fleece, or parka
	color	Attribute	Dominant color of the jacket (e.g., black, gray, green)

Table 6: Hyper-parameter settings for training the recommendation models.

Dataset	Hyper-parameters	MF	NeuMF	NGCF	LR-GCCF	LightGCN	SimGCL	MGCCF	GTN	CAGED
Kuairec	Batch Size	4096	4096	4096	4096	4096	4096	4096	4096	4096
	Layer Number	2	2	2	2	2	2	2	2	2
	Dimension Size	256	256	256	256	256	256	256	256	256
	Learning Rate	1×10^{-4}	5×10^{-7}	5×10^{-5}	1×10^{-5}	1×10^{-4}	1×10^{-4}	1×10^{-6}	1×10^{-4}	5×10^{-4}
	Regularization Rate	1×10^{-3}	0	5×10^{-3}	5×10^{-3}	1×10^{-3}	5×10^{-3}	5×10^{-3}	7×10^{-3}	5×10^{-3}
Coat	Batch Size	512	512	512	512	512	512	512	512	512
	Layer Number	2	2	2	2	2	2	2	2	2
	Dimension Size	256	256	256	256	256	256	256	256	256
	Learning Rate	5×10^{-4}	1×10^{-5}	5×10^{-5}	1×10^{-5}	1×10^{-4}	1×10^{-4}	1×10^{-5}	1×10^{-4}	5×10^{-4}
	Regularization Rate	1×10^{-2}	0	5×10^{-3}	5×10^{-3}	5×10^{-3}	5×10^{-3}	5×10^{-3}	7×10^{-3}	5×10^{-3}